Weekly schedule

Week	Topics	Assessments	
1	Introduction to AI and NLP, Pre-processing Techniques	Practice quizzes (not graded)	Traditional Machine Learning (Dr Supraja)
2	Linguistic Feature Extraction (POS, NER, DP)	Practice quizzes (not graded)	
3	Term Weighting, Topic Modeling, and Dimensionality Reduction	Coding quiz #1	
4	Classification and Clustering, Evaluation Metrics, Word Embeddings	IRA/TRA #1	
5	Neural models (RNN, LSTM, GRU) and Hyperparameter Tuning	Coding quiz #2	Deep Learning (Dr Supraja)
6	Transformers and Transformer-based Large Language Models	IRA/TRA #2	
7	NLP Applications in the Industry: Exploring the Landscape	Coding quiz #3*	Business perspective (Dr Simon)
RECESS WEEK: NO CLASS			
8	**Mid-term test covering topics from Week 1-6 (FRIDAY 9 OCTOBER) – invigilated by Dr Simon and TAs**		
9	Project briefing and feedback on common mistakes in mid-terms	Coding quiz #4*	Dr Supraja
10	Recent emerging trends (Part 1)	IRA/TRA #3^	New Topics (Dr Simon)
11	Recent emerging trends (Part 2)	IRA/TRA #4^	
12	Project feedback/consultations	Make-up quizzes	Dr Supraja & Dr Simon
13	Industry sharing	Group project due	Invited speakers

**tests only one previous week's content (Week 6 for Coding quiz #3 and Week 7 for Coding quiz #4)*

^tests only one current week's content (Week 10 for IRA/TRA #3 and Week 11 for IRA/TRA #4)

rest of quizzes will test that week and the week before

Assessment Methods

[IMPORTANT] Absence record portal (link in NTULearn) for MC/LOA/official letter

Component	Weeks	Weighting	Team/Individual
Theory Quiz (IRA and TRA)	4,6,10,11	20%	Individual readiness assessment (15%) Team readiness assessment (5%)
Mid-term Test	8 (Fri 9 October)	35%	Individual
Coding Quiz	3,5,7,9	20%	Individual
Final Capstone Group Project	13	25%	Team
Peer Evaluation	13 (via Peerceptive NTULearn)	0% (but grade deduction of 5% if did not do it)	Team
Total		100%	Individual (70%), Team (30%)

Lesson plan (weeks 1 to 7) – 2 hours

Activity	Duration
Short lecture introducing a new concept	15-20 mins
Quiz (theory or coding – depending on the assigned week)	15-30 mins
Team-based guiding questions for idea generation, reading articles, and/or writing codes	30-40 mins
Sharing (class-level)	20-30 mins

Second half

Week	Duration
8	2 hours (mid-term test, no lesson)
9	2 hours (project briefing and mid-term review)
10	3 hours (in-person lecturing and IRA/TRA)
11	3 hours (in-person lecturing and IRA/TRA)
12	3 hours (make-up quizzes and consultations)
13	3 hours (guest speakers, panel session)

Sample IRA/TRA and coding quizzes

- Closed-book tests (Access code will be provided in class)
- Using **Respondus lockdown browser** via NTULearn (attendance will be taken)
- IRA/TRA (20 mins in total) – tests content for **that week and week before + new short lecture in class** (**no need to memorize, just understand concepts from lecture video and slides, no need extra outside references**)
  - 4 quizzes in total
  - **Multiple correct answers** for every question
  - 5 questions, 10 minutes
  - Incorrect answers have a **penalty** (subtraction of marks, but **minimum is zero**)
  - First done individually (questions and options randomized order), then discuss together for the same questions and submit answers as a team

Access code for practice IRA: 549608

- Coding (15 mins) – tests content for **that week and week before + new short lecture in class** (application-based)
  - 4 quizzes in total
  - **Single correct answer** per question
  - 5 questions, 15 minutes
  - Individual only – questions and options randomized order
  - Zero marks for near-miss answers
  - **Subtraction of marks** for completely incorrect answers, can have **overall negative score**
  - **Final score will be scaled and adjusted using formula (((score*2)+20)/120)*100 to achieve minimum marks as zero**

    e.g., if minimum score is -10/50 → converted to (((-10*2)+20)/120)*100 = 0/100

    e.g., if your score is 0/50 → converted to (((0*2)+20)/120)*100 = 16.67/100

    e.g., if your score is 20/50 → converted to (((20*2)+20)/120)*100 = 50/100

    e.g., if your score is 38/50 → converted to (((38*2)+20)/120)*100 = 80/100

Access code for practice coding exercise: 316882

First team-based discussion: let's use Padlet!

Before we begin: something new this academic year → bonus participation points for teams that are active during class (included within the team assessment components)

- Discussion on data pre-processing techniques presented in the lecture (each set of groups can work on one question and write simple Python codes to demonstrate if possible)
  - **Task A**: How do you compare stemming and lemmatization (and for which context)?
  - **Task B**: What is the impact with or without the removal of stopwords on downstream tasks?
  - **Task C**: Where is regular expression mostly used and in which context specifically?
  - **Task D**: Which is the most important pre-processing technique and why?

Homework: please watch Week 2 lecture videos and try out the Jupyter notebook exercises, we will do another round of practice quizzes in class